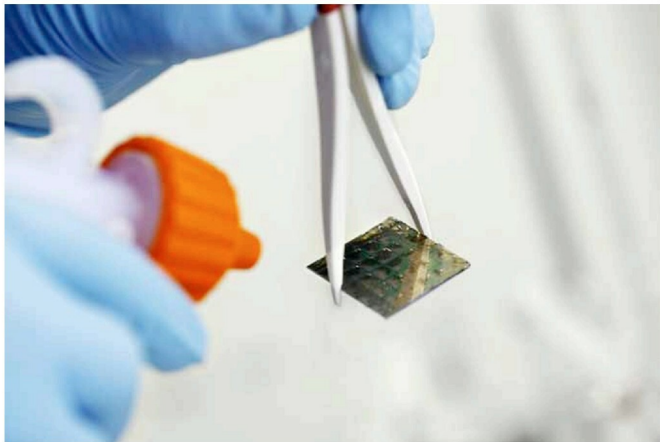


Latest in Chip Fabrication

A 'triple threat' of new complex technology advancements faces semiconductor manufacturing—three-dimensional interconnect (3D IC), extreme-ultraviolet lithography (EUV) and 450mm wafers. To reap the benefits, the industry needs to resolve both technical and business related issues



UMA GUPTA

WITH INPUTS FROM DEBASISH PAUL CHOUDHURY, SEMI INDIA PRESIDENT, AND IVO BOLSENS, SENIOR VICE PRESIDENT AND CTO, XILINX

Historically, ICs have followed Moore's Law, doubling logic capacity every two years while costing half as much with each new generation of process technology. Today, Moore's Law is suffering from mounting die costs, while no longer delivering traditional performance increases

or keeping pace with the growing demand for increased bandwidth and capacity.

The latest technologies and approaches have set out to achieve 'more than Moore' to enable next-generation silicon scaling at the current-generation process technology. The process of making chips smaller and smaller is becoming increasingly complex, utilising the most advanced capabilities of physics, chemistry, electrical and mechanical engineering among other disciplines.

Every aspect of chip manufacturing

technology is continually improved to manufacture smaller devices with better performance. Most semiconductor manufacturers and their equipment and materials suppliers spend a significant portion of their revenues on research and development.

New technology advancements

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